MOSAM BISWAS

The complete technical anatomy of *www.mosambiswas.com* — a dark editorial portfolio built with **zero frameworks and zero build step**. Version six sets the whole site on fire: a hand-written WebGL ember field, an inertial motion system, and three page-specific choreographies — all in vanilla JavaScript.

(A) — STACK	(B) — HOST	(C) — DEPENDENCIES	(D) — EDITION
HTML · CSS · Vanilla JS · GLSL	GitHub Pages + CNAME	None. Hand-rolled.	Vol. 06 — July 2026

(01) — OVERVIEW

ONE REPO, NO MACHINERY

The site is a single static repository — `BiswasMosam.github.io` — served straight from GitHub Pages at **www.mosambiswas.com**. There is no bundler, no transpiler, no package.json. Every line that ships is a line that was written by hand: `index.html`, one stylesheet, and five small scripts.

What's new in Vol. 06. Three motion engines join the v5 base: `shader.js` (a full-page WebGL ember field), `motion.js` (inertial scroll, velocity shear, letter cascades), and two page-specific systems — `cert-motion.js` (the certificates "dossier") and `sheichobi/darkroom.js` (photography "darkroom"). The hero label now reads *Portfolio — Vol. 06* and whispers: *"the smoke is hand-written GLSL."*

FILE	ROLE
`main.js`	Preloader gate, cursor/bubbles, reveals, parallax, clock, menu, modal
`motion.js`	Inertial scroll, shear, cascades, marquee, magnetic buttons — new in v6
`shader.js`	WebGL ember field, GLSL fragment shader — new in v6
`cert-motion.js`	Dossier motion for certificates.html — new in v6
`darkroom.js`	Darkroom motion for SheiChobi gallery — new in v6
`service-worker.js`	PWA cache, production-only in v6

"No frameworks, no build step — every line that ships was written by hand."

(02) — DESIGN SYSTEM

WARM BLACK, BONE INK, VERMILION

The palette is deliberately narrow: a warm near-black canvas, an off-white "bone" ink, and exactly **one accent**. Hierarchy is carried by two dimmed alpha steps of the ink rather than extra hues. Hairlines at 14% ink alpha do all the structural work — the same rule that frames this page.

The editorial voice comes from numbered section labels — `(01) — SELECTED WORK` — set in Space Mono, and from *emphasised words* dropped into Instrument Serif italic in accent colour, mid-sentence, exactly as on this page.

TOKEN	VALUE	USE
--bg	#0b0b0a	Warm black canvas
--bg-2	#131311	Raised surfaces
--ink	#edebe4	Primary text
--ink-dim	ink @ 58%	Body copy
--ink-faint	ink @ 34%	Meta text
--line	ink @ 14%	Hairline rules
--accent	#ff5227	Vermilion — the only colour

House easing: `cubic-bezier(0.22, 1, 0.36, 1)` — a long decelerating tail used by nearly every transition. Fluid padding: `clamp(1.25rem, 4vw, 4rem)`.

(03) — TYPOGRAPHY

FOUR VOICES, ONE CONVERSATION

All four families load from Google Fonts with `display=swap`; this document embeds the same faces so what you read here is what the site renders. Each face has a strict, non-overlapping job.

FACE	WEIGHTS	ROLE ON THE SITE
SYNE	600 / 700 / 800	Display. Hero name, section titles, project names — always uppercase, tracked tight at −0.02em. In v6 the display type is split into letters and animated per-glyph.
Instrument Serif	400 italic	The accent voice. Emphasised words inside headlines, the hidden "secret layer" texts revealed by the lens, pull-quotes. Nearly always paired with vermilion.
Manrope	400 / 500 / 600	Body. Paragraphs, descriptions, UI copy at line-height 1.6. Neutral on purpose — it lets the other three voices carry the character.
Space Mono	400 / 700	The label voice. Section numbers, timestamps, whisper pills, the live IST clock — 0.72rem, uppercase, letter-spaced 0.14em. Machine annotations on an editorial page.

"The type doesn't sit on the page — it arrives."

EDGE-TO-EDGE HERO FITTING — REFINED IN V6

The hero name **MOSAM / BISWAS** is fitted to the exact container width by JavaScript: each line is set at a 100px probe size, measured, then scaled by `containerWidth / wordWidth`. The second line is outlined via `-webkit-text-stroke` with a transparent fill.

v6 refinement: because pixel-based letter-spacing and strokes don't scale with font-size, one pass always lands slightly wide. `fitLines()` now runs up to **three corrective passes**, converging within 1px of the true edge.

```
let size = 100;
for (let pass = 0; pass < 3; pass++) {
  const width =
word.getBoundingClientRect().width;
  if (width <= 0) break;
  size *= line.clientWidth / width;
  line.style.fontSize = `${size.toFixed(2)}px`;
  if (Math.abs(width - line.clientWidth) < 1)
break;
}
// motion.js re-fits after the letter split:
window.__refitHero = fitLines;
```

(04) — THE EMBER FIELD

HAND-WRITTEN GLSL, NO LIBRARIES

Behind every page now sits a fixed `<canvas>` at `z-index: -1` running a raw WebGL fragment shader — no Three.js, no helper library, just a compiled program and a **fullscreen triangle** (three vertices, cheaper than a quad). The body background went transparent so the field shows through everything.

The image is **domain-warped smoke**: a hash-based value noise is stacked into four octaves of fBm, then fed into itself twice — `fbm(p + 2.1 · r)` where `r` is built from `q`, which is built from `p`. Bone-coloured smoke rises from the folds; a `smoothstep` ridge extractor paints **vermilion embers** along the crests; a vignette pulls the edges back to editorial black; a one-bit dither kills gradient banding.

```
// domain-warped fbm — the smoke
vec2 q = vec2(fbm(p*1.7 + t),
              fbm(p*1.7 - t*0.8 + 4.7));
vec2 r = vec2(fbm(p*1.7 + 2.3*q + ... + t*0.55),
              fbm(p*1.7 + 2.3*q + ... - t*0.4));
// the field leans toward the cursor
r -= (p - m) * exp(-md * 2.6) * 0.4;
float f = fbm(p*1.7 + 2.1*r);

// bone smoke + vermilion embers
col += vec3(.93,.92,.89) * f*f*f * 0.13
       * (0.35 + 0.65 * energy);
float ridge = smoothstep(0.50, 0.86,
       f * (0.55 + 0.45 * q.x));
col += vec3(1.0, 0.32, 0.15) * ridge
       * 0.42 * energy;
```

THE ENERGY SYSTEM

The field *breathes with the page*. A single `uEnergy` uniform blends four inputs: hero presence (strong at the top), contact flare (strong at **LET'S TALK**), smoothed scroll velocity, and a `uBurst` ignition that fires at 1.4 when the preloader lifts and decays by ×0.982 per frame. Mid-read, the field goes quiet.

CURSOR WARMTH

The mouse position is lerped (0.07) into a `uMouse` uniform; a Gaussian falloff `exp(-d²·7)` adds local heat, scaled by smoothed cursor speed. The warp itself is bent toward the pointer, so the smoke *leans into your hand*.

PERFORMANCE DISCIPLINE

The canvas renders at **0.5× resolution** (capped at 1.5 dpr) — the field is soft by nature, so nobody can tell. Context flags: `alpha:false, antialias:false, depth:false, powerPreference:'low-power'`. An **adaptive quality governor** counts frames over 27ms; after 45 slow frames it degrades resolution once more by ×0.65, permanently. Under `prefers-reduced-motion`, or if WebGL/compilation fails, the canvas removes itself and the site falls back to plain `--bg` — the shader is decoration, never a dependency.

(05) — THE MOTION SYSTEM

THE PAGE LEANS INTO ITS OWN MOMENTUM

INERTIAL SCROLL

On fine-pointer devices, `wheel` events are intercepted (`preventDefault`) and fed into a target value; a rAF loop eases the real scroll position toward it at **lerp 0.11**. Native jumps — keyboard, scrollbar drags — are detected when the actual position drifts >1.5px from the last write, and the system *resyncs instead of fighting*. Anchor links glide on the same easing. `html.smooth` disables native smooth-scroll so the two never conflict.

VELOCITY SHEAR

Scroll velocity is smoothed (lerp 0.12) and mapped to a `skewY` of at most ±3.4° applied to the hero, section heads, work list, research and photography blocks — the page visibly *shears with momentum* and settles when you stop.

LETTER & WORD CASCADES

The hero and contact titles are split into per-letter `.hero__ltr` spans that rise from `translateY(118%) rotate(5deg)` inside the existing overflow masks, staggered **50ms per letter** via a `--i` custom property. Section titles split into words (70ms stagger); work rows and photo strips enter with 60/90ms staggers under IntersectionObserver.

VELOCITY-DRIVEN MARQUEE

The CSS keyframe marquee hands over to JS (`.is-js`): base drift 42px/s, plus a velocity term clamped at ±860 — it **accelerates as you scroll and reverses when you backtrack**, wrapping by modulo on the half-width of the duplicated track.

THE REST OF THE CHOREOGRAPHY

Header hides past 280px when velocity > 5, returns on any upward motion.

Progress bar — a 2px vermilion hairline scaled `scaleX(y / maxScroll)`.

Hero lines drift apart — MOSAM slides left, BISWAS right, up to 220px as you leave the hero.

Magnetic buttons — CTAs and the menu button lean toward the cursor (0.26x / 0.34y of the offset).

Stat counters — about-section numbers count up over 1.2s with cubic ease-out, zero-padded.

Scroll hint fades out after 80px.

Everything is gated behind `html.has-motion`, which `motion.js` adds only when `prefers-reduced-motion` is off — with JS disabled or motion reduced, the stylesheet's v6 block is inert and the v5 reveals still work.

"Hand-rolled, no libraries" — the header comment of every v6 file.

(06) — BUBBLES

A CURSOR THAT WHISPERS & REVEALS

The signature interaction. One fixed `#cursor` div — a 12px white dot in `mix-blend-mode: difference` — lerp-follows the pointer (factor 0.22) and morphs into three distinct instruments depending on what it touches. Desktop only (`pointer: fine`), removed entirely under reduced motion.

WHISPER MODE — REACTING TO SEPARATE WORDS

Any element carrying `data-whisper` morphs the dot into an ink-coloured **caption pill** with Space Mono text. Crucially, whispers are wired to *individual words*: in "Developer, researcher, photographer" each word owns its own whisper, so sweeping the sentence produces a running commentary. A `||` delimiter stores alternate lines that **cycle on every re-hover** (tracked via `data-whisperIdx`).

The pill drifts 26px above the pointer, flips below near the viewport top (`targetY < 60 → +34`), clamps at the horizontal edges, and sizes itself to `label width + 36px × 32px`. Links grow the dot to 52px instead — unless they whisper too.

v6 additions: the version label whispers *"the smoke is hand-written GLSL"* / *"no libraries — just math and embers"* — the site annotating its own shader.

LENS MODE — THE READING LENS

Paragraphs marked `.secret` carry two layers: the visible `.secret__base` and an absolutely-positioned `.secret__layer` — alternate text set in accent-coloured Instrument Serif italic. Over a secret, the dot expands into a **78px-radius transparent circle** with a vermilion border, and the hidden layer is unmasked through a per-frame `clip-path: circle(r at x y)` that tracks the cursor. Move the lens, read the subtext.

```
layer.style.clipPath =
  `circle(${r}px at ${x}px ${y}px)`;
// lens radius lerp 0.16 · pill offset 0.2
```

In v6 the bubbles system is unchanged at its core but now coexists with the ember field: the same lerped cursor position that drives the pill also feeds the shader's `uMouse`, so the smoke and the bubble move as one hand.

(07) — PRELOADER & LIFECYCLE

HELD AT 99 UNTIL IT'S TRUE

The v5 preloader counted 00→100 on a fixed one-second cubic ease and lifted its curtain regardless of what had actually loaded. **In v6 the count is a real load gate:** it holds at **99** until `document.fonts.ready` and the window `load` event have both resolved, so the curtain never lifts onto a page that still shifts. A 3.5s `MAX_WAIT` cap races the gate so slow networks can't trap anyone.

```
Promise.race([
  Promise.all([fontsReady, pageLoaded]),
  cap  // setTimeout(resolve, 3500)
]).then(() => { assetsReady = true; });

if (!assetsReady) count = Math.min(count, 99);
```

When the curtain lifts (`translateY(-100%)`), `body.is-loaded` triggers the hero letter cascade — and the shader's ignition **burst** fires at the same moment. Load, ignite, cascade: one choreographed instant.

SERVICE WORKER — PRODUCTION ONLY

New in v6: the worker registers only when the hostname matches `mosambiswas.com`. On any dev origin, previously registered workers are **unregistered and all caches deleted**, so localhost can never serve stale files. In production it keeps the cache-first strategy with a `SKIP_WAITING` update flow and `updateViaCache: 'none'`.

KEPT FROM V5

Film grain — a fixed SVG `feTurbulence` data-URI at 0.045 opacity over everything.

Scroll parallax — `data-parallax` images translate and scale (1.12) against scroll.

Work-list hover — rows invert via `scaleY` ::before wipe; a cursor-following preview card (lerp 0.12) strips to the project image by `index × -100%`.

Live IST clock — `Intl.DateTimeFormat` in Asia/Kolkata, ticking every second.

Copy-email with Clipboard API + `execCommand` fallback; fullscreen mobile menu with staggered links; right-click guard; console signature; ASCII art in the source.

(08) — CERTIFICATES: THE DOSSIER

PROOF OF WORK, DECLASSIFIED

The certificates page gets its own motion language — *paper, stamps, and filing cabinets* — gated behind `html.has-dossier`.

REDACTION-BAR HERO

The title **PROOF OF WORK** arrives like a declassified document: solid bars — ink for the first line, vermilion for the outlined second — **sweep across each line** (scaleX 0→1 from the left, then 1→0 from the right), and the text snaps visible mid-sweep, exactly when the bar covers it.

```
@keyframes dossier-sweep {
  0%   { transform: scaleX(0); origin: 0 50%; }
  46%  { transform: scaleX(1); origin: 0 50%; }
  54%  { transform: scaleX(1); origin: 100%
50%; }
  100% { transform: scaleX(0); origin: 100%
50%; }
}
```

STAMPED FEATURES

Featured certificates land with an overshoot bezier `(0.2, 0.9, 0.3, 1.15)` from `scale(1.05) rotate(0.6deg)` — stamped onto the page — followed by a decaying vermilion **stamp-flash** box-shadow. The modal image reuses the same stamp-in.

FILED TILES & TYPED LABELS

Archive tiles file in one by one — 55ms stagger, each with a slight index-derived rotation `(--i - 3) × 0.35deg`, like papers squared off by hand. Section labels **type themselves** character-by-character with a ▌ caret at 26ms per glyph; the "08 certificates" aside counts itself up. All triggered by IntersectionObservers.

PAPER TILT + SHEEN

On desktop, every certificate tilts in 3D under the cursor — `perspective(720px)` with up to 9°/7° of rotation — while a radial **sheen** follows the pointer via `--gx`/`--gy` custom properties, like light raking across a glossy print.

```
el.style.setProperty('--ry',
  `${((px - 0.5) * 9).toFixed(2)}deg`);
el.style.setProperty('--rx',
  `${((0.5 - py) * 7).toFixed(2)}deg`);
```

(09) — SHEICHOBI: THE DARKROOM

PRINTS THAT DEVELOP AS YOU WATCH

Shei Chobi — Bengali for "that picture" — is the photography wing, and in v6 it behaves like a darkroom: *light, chemistry, and optics*, gated behind `html.has-darkroom`.

THE DEVELOPING BATH

The hero title **SHEI CHOBI** arrives overexposed — `blur(18px) brightness(2.3)` — and settles into focus over 1.7s, like a print in developer fluid. Every archive photo goes through the same bath: from `blur(14px) brightness(1.9) grayscale(1)` to full colour, batch-staggered 90ms. Because the grid is rebuilt on every filter and load-more, a **MutationObserver** re-watches new cards automatically.

FOCUS PULL

Hovering the contact sheet racks focus like a lens: the hovered frame stays sharp while every sibling falls to `blur(3px) brightness(0.75)` — a pure-CSS `:hover` chain, desktop only.

APERTURE LIGHTBOX

Clicking a photo opens the lightbox as an **aperture**: a `clip-path: circle()` expands from 0% to 141% *centred on the exact click point*, captured into `--cx`/`--cy` custom properties. Between photos, a **shutter blink** — brightness flash and blur, restarted by forcing reflow when the image `src` mutates.

```
lightbox.style.setProperty('--cx',
  `${e.clientX}px`);
clip-path: circle(141% at var(--cx) var(--cy));
```

BREATHING GRAIN

A rAF loop smooths scroll velocity and applies a **motion blur** to the sheet (up to 2.6px) while the film-grain overlay's opacity breathes with speed — scroll fast and the grain thickens, stop and it settles back to 0.045.

"Every page speaks the same design language, but each one moves in its own dialect — embers, dossiers, darkrooms."

(10) — ENGINEERING LEDGER

FAST, ACCESSIBLE, HONEST

PERFORMANCE DECISIONS

Zero dependencies — no framework tax; the entire v6 motion stack is ~660 lines of vanilla JS.

Transform/opacity only — every animation runs on compositor-friendly properties; staggers via `--i` custom properties, not per-element JS timers.

One rAF loop per system — motion, shader, darkroom each run a single frame loop; observers do the triggering.

Adaptive shader quality — 0.5× render scale, low-power context, self-degrading on slow frames.

Preloader as load gate — no layout shift after the curtain.

ACCESSIBILITY & RESILIENCE

Reduced motion — all four v6 engines bail on `prefers-reduced-motion`; the shader canvas removes itself; CSS gates (`.has-motion`, `.has-dossier`, `.has-darkroom`) make v6 styles inert without JS.

No-JS fallback — reveals gated behind `html.js`; content is fully readable without a single script.

Skip-link, sr-only h1, semantic landmarks, keyboard-reachable modals.

SEO & PWA

JSON-LD — WebSite + Person structured data; OG/Twitter cards; sitemap + robots.txt.

PWA — manifest + service worker, production-gated in v6 with dev-origin cache eviction.

Canonical redirects for the gallery; CNAME to www.mosambiswas.com.

CONTENTS OF RECORD

Nine projects — Sim Sim, Rail Mitra, MindFirst, PixelForge, IntelliStatement, Committee Connect, PixelShift, Flex Card, To-Do.

IEEE publication — psychometric ML screening, ICAIIHI 2025, DOI 10.1109/ICAIIHI67124.2025.11403838.

SheiChobi photography gallery · certificates archive · résumé.

(●) — CONTACT

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